EXPERIMENT 4: STUDY THE RESPONSE OF A SAMPLE-AND-HOLD (S/H) CIRCUIT

OBJECTIVE:

1. To study the response of a Sample-and-Hold (S/H) circuit.

2. To study the pin configuration of 555 Timer IC and Op-Amp (LM741) IC and their uses in S/H circuits.

INTRODUCTION:

The **Sample-and-Hold (S/H) circuit** is used to sample an analog input voltage for a very short period (microseconds to milliseconds) and to hold its last sampled value until the input is sampled again. The period may range from a few milliseconds to several seconds.

The S/H circuit has a **third terminal for the logic S/H command** in addition to the input Vs and output Vo. This command is issued as a pulse to control the sampling process.

The circuit operates in **two modes** depending on the logic level of the S/H pulse:

- **Track Mode:** During the S/H pulse, Vo follows Vs.

- **Hold Mode:** After the pulse, Vo is held at the value of Vs at the trailing edge of the pulse.

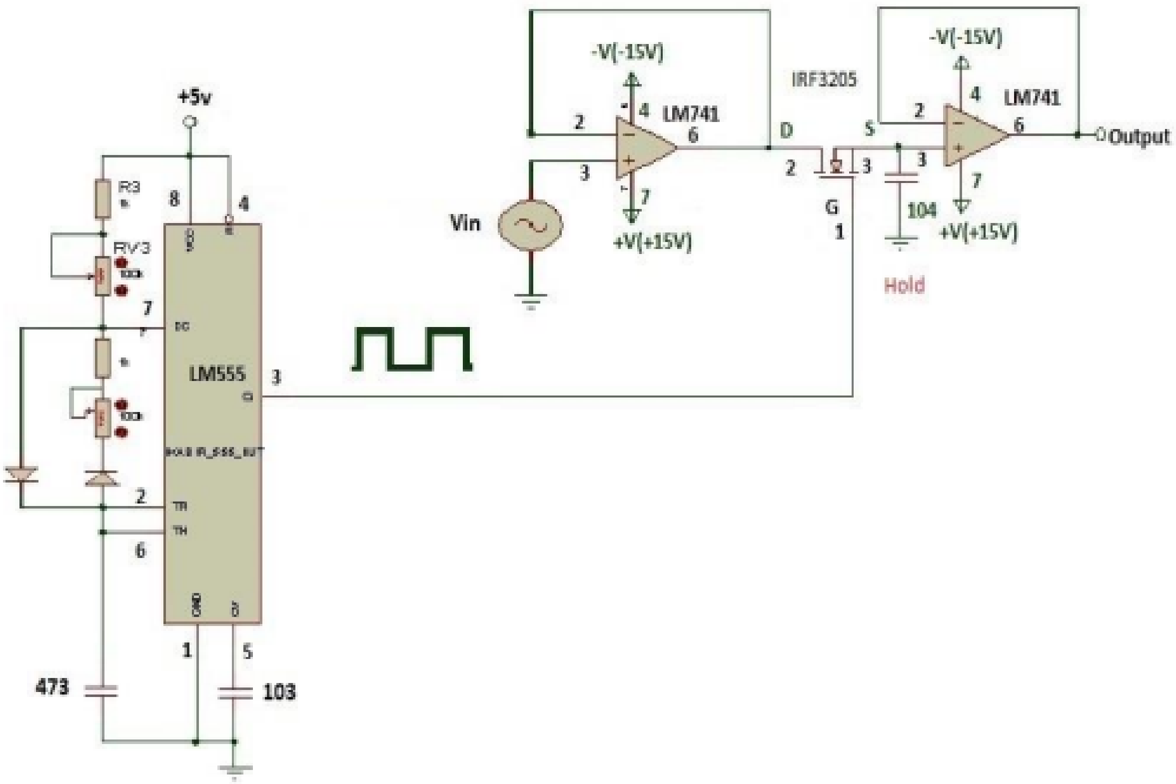
A **periodic pulse train** allows the entire analog signal to be sampled, producing a discrete version of the signal at the output. To avoid aliasing, the **sampling frequency** should be at least twice the bandwidth of the input analog signal.

Sample-and-Hold circuits are used to hold signals constant for A/D converters, remove glitches in D/A converter outputs, and enable fast acquisition with minimal voltage droop.

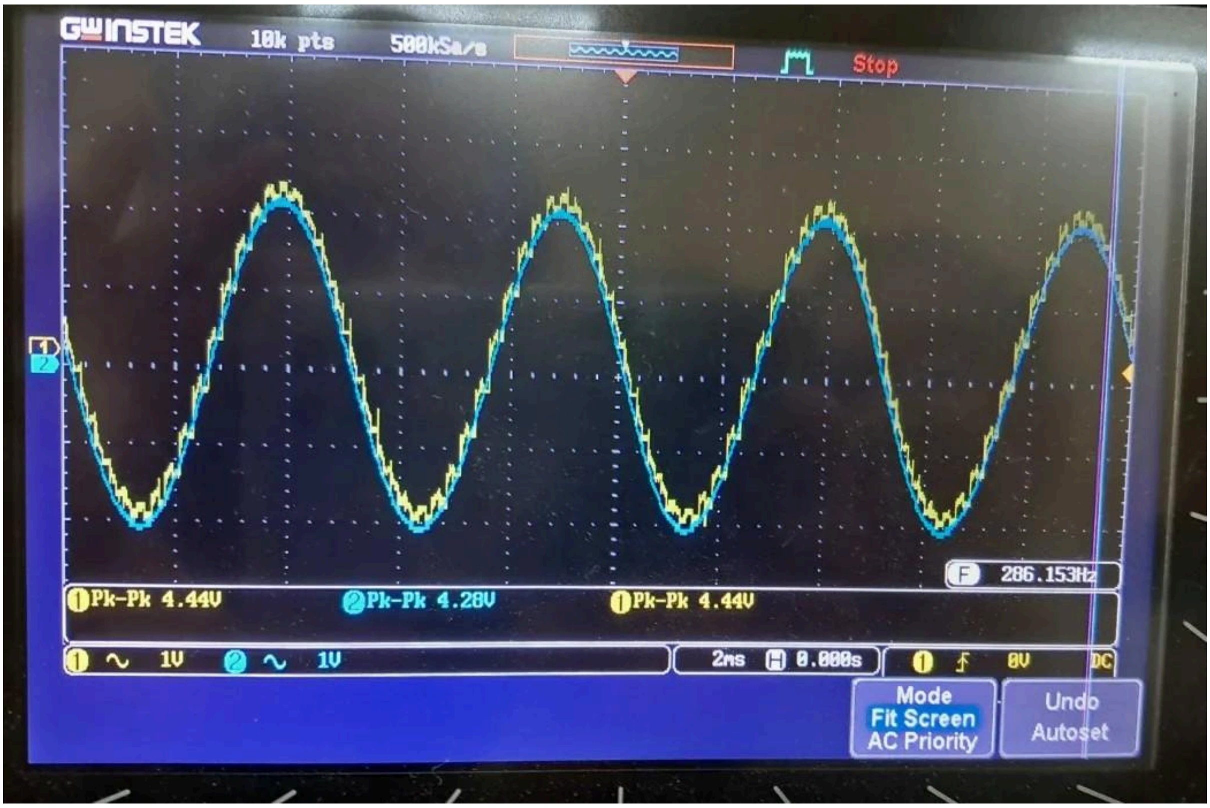
LIST OF EQUIPMENT:

1. DC Power Supply,
2. Function/Signal Generator,
3. Breadboard,
4. Resistor-1 kΩ (2 Nos), 100 kΩ (2 Nos),
5. Potentiometer-10 kΩ (2 Nos),
6. Diode (2 Nos),
7. Capacitor-0.01 μF (2 Nos), 0.047 μF (2 Nos),
8. 555 Timer IC,
9. Op-Amp (LM 741),
10. MOSFET IFR-540,
11. Oscilloscope, and
12. Connecting wires.

CIRCUIT DIAGRAM for Basic S/H Circuit Configuration (Fig. 4c):



Graph:



JUSTIFICATION / EXPLANATION OF THE EXPERIMENTAL DATA

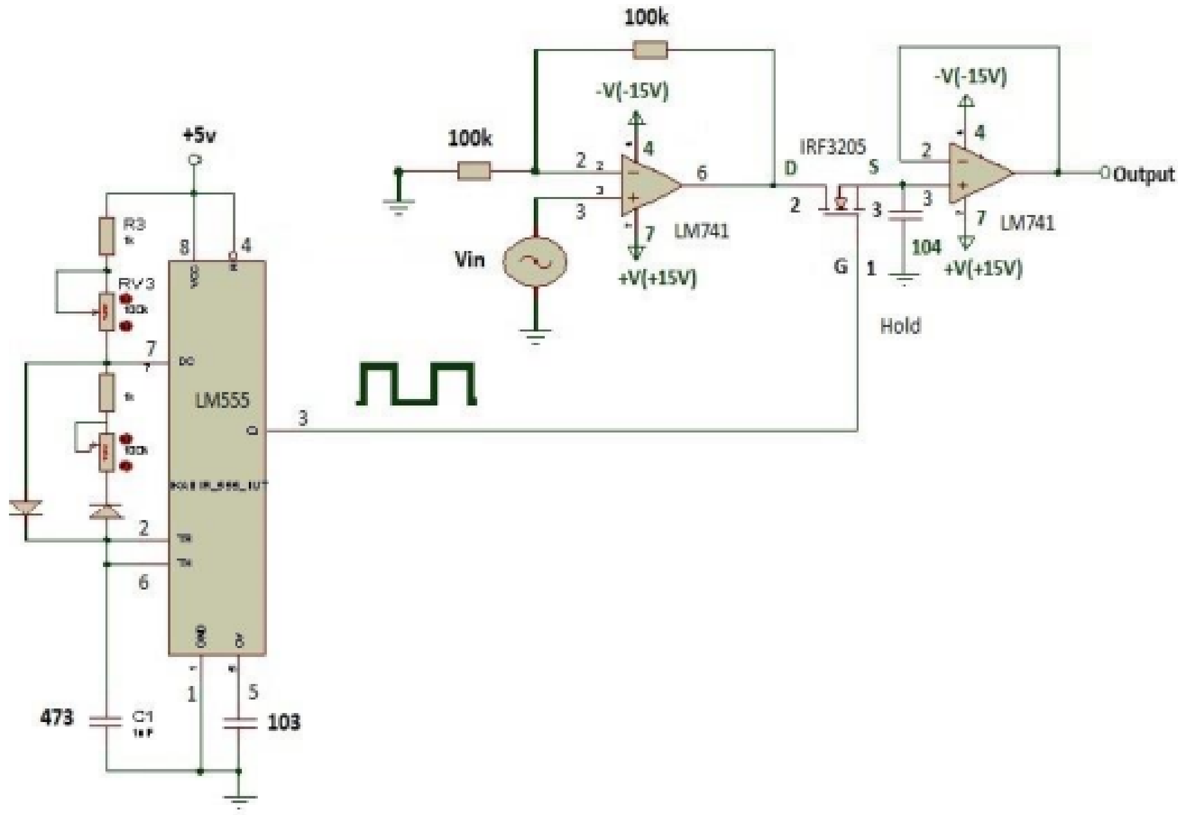
Concept:

- During the sampling pulse, the MOSFET charges the holding capacitor to the input voltage.
- After the pulse ends, the MOSFET turns off, and the voltage across is maintained by the second Op-Amp, preventing rapid discharge.
- Acquisition time depends on the RC constant of the MOSFET and capacitor, the output current capability of the first Op-Amp, and its slew rate.

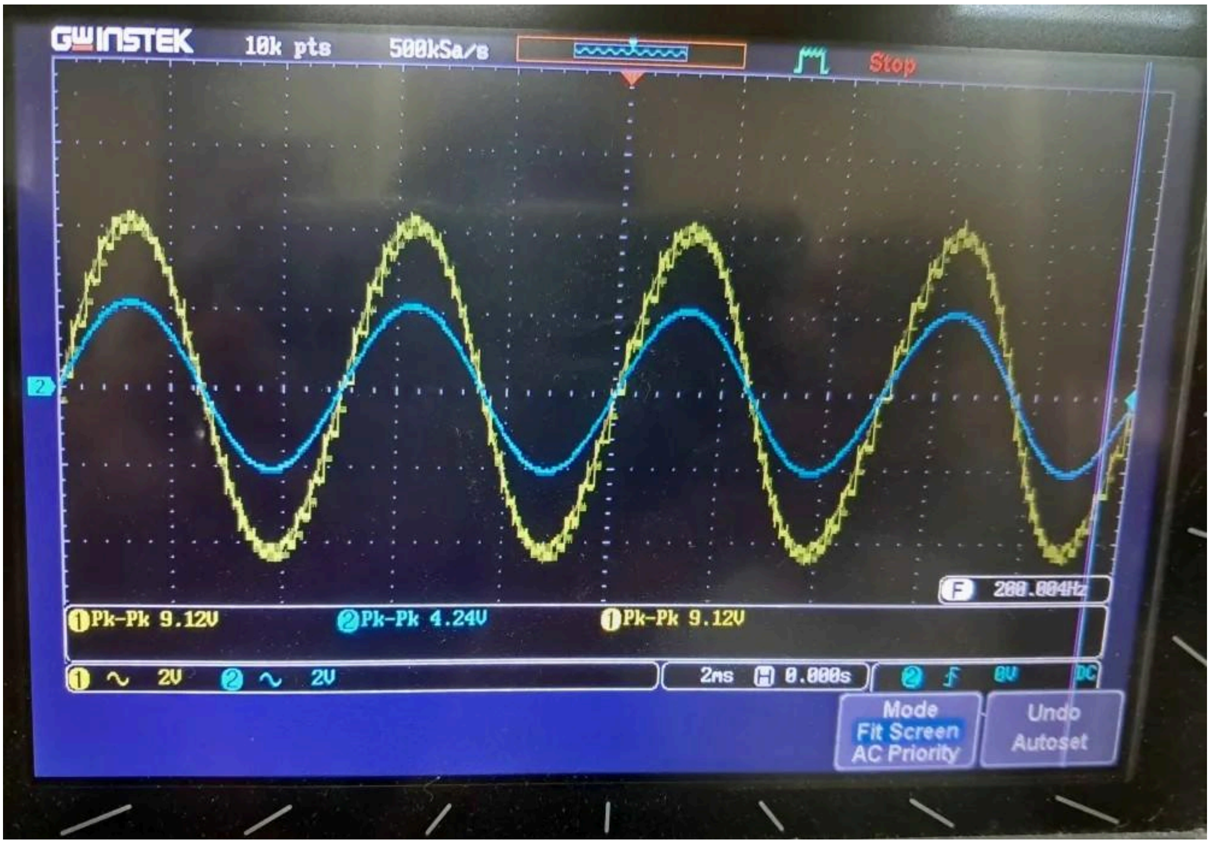
Experimental Results:

- Output follows the input during sampling and holds steady during the hold period.
- Higher input frequencies require faster acquisition for accurate tracking.
- Varying the duty cycle changes acquisition and hold times.
- Increasing the holding capacitor reduces voltage droop but slightly increases acquisition time.

CIRCUIT DIAGRAM for **Basic S/H Circuit with Gain (Fig. 4d):**



Graph:



JUSTIFICATION / EXPLANATION OF THE EXPERIMENTAL DATA

Concept:

- Similar operation as the basic circuit, but the second Op-Amp is configured with feedback to provide voltage gain.
- The amplified output improves acquisition speed and hold stability while driving loads more effectively.

Experimental Results:

- Output amplitude is higher due to gain.
- Acquisition time is slightly faster, and the held voltage is more stable.
- Effects of input frequency and duty cycle are similar but better compensated due to amplification.
- Larger holding capacitor still reduces droop but slightly slows acquisition, similar to the basic circuit.

ANSWERS TO REPORT QUESTIONS:

1. **Differences between Fig. 4(c) and Fig. 4(d):**

- **Fig. 4(c):** This is the basic Sample-and-Hold (S/H) circuit. It samples the input signal using a MOSFET switch and stores the voltage on a holding capacitor, which is then buffered by an Op-Amp to maintain the held value.

- **Fig. 4(d):** This circuit modifies the basic configuration by adding Op-Amp feedback in the output stage to provide voltage gain. This not only increases the output amplitude but also improves acquisition time and enhances hold stability by better isolating the capacitor from the load.

2. **Response to Sampling Frequency and Duty Cycle:**

- **Sampling Frequency:** At higher sampling frequencies, the circuit can track the input more accurately because the capacitor is updated more often. At lower frequencies, the hold period becomes longer, leading to increased voltage droop and less accurate reproduction of the input signal.

- **Duty Cycle:** A longer duty cycle allows more time for acquisition, enabling the capacitor to charge closer to the input value, but reduces the hold period. A shorter duty cycle results in quicker sampling with a longer hold period, but may cause incomplete charging of the capacitor, leading to reduced accuracy in the sampled value.

DISCUSSION:

This experiment demonstrated the Sample-and-Hold circuit using MOSFET switches and Op-Amps. The effect of input frequency, duty cycle, and capacitor value on acquisition and hold was observed. The addition of gain through feedback improved output stability and acquisition speed. The experiment reinforced understanding of analog sampling, circuit timing, and the practical use of S/H circuits in A/D and D/A systems.